KASANDY FOUNDATION

Kasandy Circular Economy Initiative

RESEARCH AGENDA & FEASIBILITY ASSESSMENT

Textile-Based Insulation Panels: Feasibility Assessment & 2026–2028 Research Agenda

Assessment of technical and regulatory feasibility, with a structured research agenda for pursuing insulation as a medium-term KCEI product stream.

Kasandy Circular Economy Initiative · ECUAD Partnership · 2026

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Insulation panels are a research target for 2026–2028, not a committed product stream. Commercialization is contingent on National Building Code compliance.

FEASIBILITY ASSESSMENT & RESEARCH AGENDA

Textile-Based Insulation Panels: Feasibility Assessment & 2026–2028 Research Agenda

This document serves two purposes: (1) an honest assessment of the technical and regulatory feasibility of producing construction insulation panels from post-consumer textile waste, and (2) a structured research agenda for pursuing insulation as a medium-term KCEI product target through 2028.

The short version: textile-based insulation is technically feasible and the material properties of textile fibres make them well-suited to insulation applications. However, the regulatory pathway to commercial deployment in Canadian construction is complex, time-consuming, and capital-intensive. KCEI will pursue insulation development systematically through the ECUAD research partnership, while maintaining acoustic bricks, playground surfaces, and park furnishings as the primary near-term commercial targets.

Assessment summary & 2026–2028 targets

- Technical feasibility: SUPPORTED — Textile fibres share key properties with commercial insulation materials
- Regulatory pathway: COMPLEX — National Building Code requires third-party certification (12–24 months, $50K–$150K)
- Market opportunity: REAL — Green building market is growing; low-embodied-carbon insulation is in demand
- 2026: Material testing under ECUAD partnership; baseline thermal, acoustic, and fire performance data
- 2027: Fire retardant formulation development; assessment of certification investment viability
- 2028: Certification pathway decision point; alternative compliance pathway exploration for specific projects

SECTION 1: TECHNICAL FEASIBILITY

Technical Feasibility

Why textile fibres are suited to insulation

Construction insulation works by trapping air within a porous matrix, slowing the transfer of heat. Natural and synthetic textile fibres are inherently porous, available in bulk compressed form, and dimensionally compatible with standard wall cavity and ceiling assembly dimensions. Natural fibres (cotton, wool) offer the additional advantage of moisture management: they absorb and release moisture without losing thermal performance, unlike fibreglass which degrades in damp conditions. These properties make textile waste a genuinely promising insulation feedstock.

Thermal performance expectations

Published research on textile-based insulation products (including cellulose fibre insulation, which is a well-established product) suggests that compressed textile batts can achieve R-values of R-2.5 to R-4.5 per inch of thickness, depending on fibre composition, binder content, and compression density. This range is comparable to standard fibreglass batt insulation (R-3.1 to R-3.7 per inch). For many interior wall applications and milder climate zones, this performance is sufficient. For high-performance building envelopes or compliance with BC Energy Step Code higher tiers, supplemental insulation would likely be required.

Fire performance — the key design challenge

Fire performance is the primary technical constraint. Natural fibres (cotton, wool) are combustible and require fire retardant treatment for use in regulated wall and ceiling assemblies. Synthetic fibres (polyester) melt rather than burn but produce toxic combustion gases. The research agenda must include fire retardant formulation development as a core workstream. A blended fibre approach with appropriate treatment is the most likely pathway to a compliant product.

SECTION 2: REGULATORY PATHWAY

National Building Code Pathway

This is the most significant constraint on insulation commercialization. The National Building Code of Canada (NBC) and BC Building Code require that insulation in regulated locations (wall cavities, ceiling assemblies, floor assemblies) carry third-party certification from a Standards Council of Canada-recognized body. For thermal insulation batts, the relevant standard is CAN/ULC-S702 or ASTM C665.

Certification investment

Achieving product certification for a novel insulation material from a small manufacturer requires: laboratory testing at an accredited facility; product documentation; quality management system audit; and ongoing certification body surveillance. The total investment is typically $50,000–$150,000 and takes 12–24 months. This investment is not appropriate for KCEI’s current pilot phase but becomes viable as the product is validated through R&D; and the commercial opportunity is confirmed.

Alternative: Section 2.5 alternative compliance

For specific construction projects, the NBC’s Section 2.5 alternative compliance pathway allows the authority having jurisdiction (AHJ) to accept alternative materials with demonstrated equivalent safety. This pathway is discretionary, project-specific, and cannot substitute for product certification in commercial or multi-family residential applications. However, it may provide an early commercial pathway for KCEI insulation in specific municipal or institutional projects where the buyer is willing to engage the AHJ approval process.

SECTION 3: 2026–2028 RESEARCH AGENDA

2026–2028 Insulation Research Agenda

The following research agenda is structured to progress insulation from concept to certification-ready product over the 2026–2028 period, in parallel with KCEI's primary product commercialization activities.

2026: Foundation year

- Produce insulation test specimens under ECUAD partnership using industrial shredding equipment (available from July 2026)
- Measure baseline thermal conductivity (lambda value) across fibre composition variables
- Preliminary acoustic and vapour permeability testing
- Literature review of fire retardant systems compatible with textile fibre matrices
- Interim findings: Is thermal performance competitive? What fire retardant approach is most promising?

2027: Performance validation year

- Fire retardant formulation development in partnership with ECUAD chemical engineering advisors
- Fire testing of treated specimens: initial classification under CAN/ULC-S102 (surface burning characteristics)
- Long-term thermal stability testing: performance under repeated humidity and temperature cycling
- Market engagement: conversations with green building architects and specifiers to assess demand and specification barriers
- Decision point: Is the certification investment viable based on performance data and market engagement?

2028: Pathway year

- If Year 2 assessment is positive: initiate certification process with accredited body; explore grant funding for certification costs
- If standard certification is deferred: develop Section 2.5 alternative compliance package for specific municipal or institutional project application
- Publish findings through ECUAD academic channels
- Refine product roadmap: insulation as Year 4+ commercial product stream

Near-Term Primary Products (2026–2027)

While insulation R&D; progresses, KCEI's commercial focus remains on products with shorter pathways to procurement-ready status:

- Acoustic bricks: Non-structural interior application; lower regulatory bar; established international proof-of-concept (FabBrick)
- Playground surfaces: Municipal procurement target; City of Vancouver, CRD, and City of Victoria engaged
- Park furnishings: Municipal procurement target; ESG-aligned corporate campus applications also viable

These products are not in competition with insulation — they serve different markets and regulatory environments. The commercial success of the near-term products will fund the longer-term insulation R&D.;

Kasandy Circular Economy Initiative. ECUAD Research Partnership. Research commencing July 2026. National Building Code references: NBC 2020, Division B, Section 2.5; CAN/ULC-S702; ASTM C665. Contact: hello@kasandyfoundation.org. kasandyfoundation.org.